Seth Robles

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Education

Massachusetts Institute of Technology (MIT) — GPA 4.8/5.0 **Expected May 2027**
BS in Mechanical Engineering and Robotics *Cambridge, MA*

- **Coursework**: Robotic Manipulation, Dynamics & Controls II, Design of Feedback Systems, Measurement & Instrumentation, Circuits, Signal Processing, Design and Manufacturing, Stochastic Systems
- **Other**: MIT Electronics Research Society, Admissions Ambassador, Camp Kesem, Latino Cultural Center Exec

Experience

Marine Autonomy Lab **Aug. 2025 – Present**
Mechanical Engineering Research Intern *Cambridge, MA*

- Integrated a Ping360 scanning sonar onto a BlueBoat USV with UDP streaming for real-time visualization and logging, supporting development of sonar-based SLAM in GPS-denied environments
- Designed and assembled a custom 12s3p Li-ion battery pack with 3D-printed structure for autonomous marine platforms deployed in low-resource environments
- Designed a pressure/temperature sensing PCB in KiCad for AUV hull-integrity monitoring, fabricated via JLCPCB
- Designed and machined sensor and communication system mounts for field deployments contributing to an OCEANS 2026 paper submission

Confidential Client **Mar. 2026 – Present**
Mechatronics Engineering Consultant *Remote*

- Primary developer of a ROS 2 autonomy stack with Gazebo simulation, supporting both MAVLink and DDS vehicle interfaces for multi-platform deployment
- Built custom Ubuntu images for Qualcomm Rubik Pi 3 compute modules to support onboard autonomy workloads

Fabrication-Integrated Design Lab **Jun. 2025 – Aug. 2025**
Mechanical Engineering Research Intern *Cambridge, MA*

- Designed and prototyped a tracked suction-based climbing robot, iterating geometry to improve adhesion reliability
- Characterized suction cup array performance through controlled load testing to establish design margins
- Developed embedded control firmware on an ATmega2560 for locomotion across varied surface geometries

Projects

Robotics Hardware Contributor **Feb. 2026 – Present**
Metal2Humanoid *Homebrew Robotics Network*

- Fabricated a working electric motor from 3D-printed components and hand-wound coils as a motor manufacturing demo
- Developing a dual-channel motor controller PCB to drive humanoid joint actuators
- Contributed to open-source educational materials on controls and motor manufacturing for humanoid robotics

Lead-Lag Control of Electromechanical Systems **Spring 2026**
2.14 Design of Feedback Systems — Labs & Design Problems *MIT*

- Designed and hardware-validated a lead-lag compensator for DC motor position control, achieving 45 rad/s crossover and 41° phase margin as verified by dynamic signal analyzer measurement
- Implemented discrete-time PID and lead compensation (Tustin, matched pole-zero) on a myRIO at 1 kHz, reaching crossover within 3% of the continuous-time design target
- Automated frequency response measurement with a Python/pyVISA pipeline driving an oscilloscope's waveform generator over SCPI, sweeping 40 log-spaced points to CSV for MATLAB analysis

Skills

Robotics: Linux (Ubuntu 24.04 Primarily), MOOS-IvP, ROS2, OpenAI Gym, Raspi, Arduino, PX4
Software & Tools: C++, Python, MATLAB, Git, Bash, Zsh
Electrical & PCB Design: KiCad, power regulation, sensor integration, STM32 Development
Mechanical Design: SolidWorks, Autodesk Fusion 360, Blender
Fabrication: Mill, Lathe, Waterjet, CNC, FDM/SLA 3D printing, Lasercutter
Languages: Spanish, Italian

Awards

Gates Scholar, Jack Kent Cooke Scholar, National Merit Scholar, QuestBridge Scholar, Hispanic Scholarship Fund Scholar